

From DFT to MD: An Introduction to Machine-Learned Interatomic Potentials

Yi-Ting Tseng

July 20, 2026

Abstract

Material properties are ultimately governed by the behaviour of atoms and electrons. Two complementary computational frameworks address this problem at different scales: Density Functional Theory (DFT) solves the electronic structure problem at quantum-mechanical precision but is limited to small systems and short time scales, while Molecular Dynamics (MD) propagates atomic positions classically and can access nano- to microsecond dynamics for millions of atoms at the cost of empirical force-field accuracy. Machine-Learned Interatomic Potentials (MLIPs) bridge this gap by encoding DFT-quality potential energy surfaces into fast machine-learning models, enabling large-scale MD simulations at near-DFT accuracy. This note introduces both methods from first principles, compares their assumptions and limitations, and explains how MLIPs connect them. Hands-on toy-model exercises are collected in the appendices.

Contents

I	Introduction	2
1	From DFT to MD	2
1.1	Overview	2
1.2	Density Functional Theory (DFT)	2
1.3	Molecular Dynamics (MD)	2
1.4	Ab Initio Molecular Dynamics (Ab Initio MD, AIMD)	3
1.5	Differences Between Methods	4
1.6	Supplement: Comparison with DFT/MD/DMFT	6
1.6.1	DFT+DMFT and AIMD: differences	6
1.6.2	Differences and limitations of DFT / MD / DMFT	6
1.6.3	Toy Model	6
1.6.4	Research story so far	6
2	Machine-Learned Interatomic Potentials (MLIP)	8
2.1	Brief History of MLIP	8
2.2	Workflow	8
2.3	Differences from Other Computational Methods	9
2.4	Active Learning for Force Fields	9
II	Exercise	10
A	Toy Model: MD with LJ/EMT	10
A.1	Lennard-Jones Potential: the simplest interatomic model	10
A.2	Physical Observables	10
A.2.1	Energy and Temperature	10
A.2.2	Radial Distribution Function (RDF)	13
B	Toy Model: MD with MACE	15
C	Next Steps in Research	16

Part I

Introduction

1 From DFT to MD

1.1 Overview

The properties of materials are, at the most fundamental level, determined by the behaviour of atoms and electrons. Computing these properties requires solving the equations of quantum mechanics, but for a many-body system this is mathematically intractable without approximation.

Density Functional Theory (DFT). Proposed by Hohenberg and Kohn in 1964, with the Kohn–Sham formulation established in 1965, DFT reformulates the many-electron problem in terms of the electron density $\rho(\mathbf{r})$ rather than the full many-body wavefunction, dramatically reducing computational complexity. DFT has successfully predicted lattice constants, band structures, phonon frequencies, and chemical reaction barriers, making it the cornerstone of modern materials computation.

Molecular Dynamics (MD). First used by Alder and Wainwright in 1957 to simulate hard-sphere systems, and extended by Rahman in 1964 to liquid argon, MD takes a fundamentally different approach: rather than solving the electronic problem, it treats nuclei as classical particles and tracks their motion via Newton’s equations. MD has successfully described liquid dynamics, protein folding, heat conduction, and diffusion in materials.

1.2 Density Functional Theory (DFT)

Starting point. Within the Born–Oppenheimer approximation the nuclear and electronic degrees of freedom decouple, and one solves only the electronic Schrödinger equation:

$$H_e \Psi = E_e \Psi, \quad H_e = - \sum_i \frac{\hbar^2 \nabla_{\mathbf{r}_i}^2}{2m_e} - \sum_{i,l} \frac{Z_l e^2}{|\mathbf{r}_i - \mathbf{R}_l|} + \sum_{j>i} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}. \quad (1)$$

Equation: Kohn–Sham equation. The Hohenberg–Kohn theorem states that the ground-state energy is a unique functional of the electron density $\rho(\mathbf{r})$. The Kohn–Sham mapping replaces the interacting many-body problem with a set of non-interacting single-particle equations that reproduce the true ground-state density:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}[\rho](\mathbf{r}) \right] \psi_k(\mathbf{r}) = \varepsilon_k \psi_k(\mathbf{r}), \quad (2)$$

with $V_{\text{eff}} = V_{\text{ext}} + V_{\text{Hartree}}[\rho] + V_{\text{XC}}[\rho]$.

Approximation. The exchange-correlation functional $V_{\text{XC}}[\rho]$ is unknown and must be approximated (LDA, GGA, etc.).

Self-consistency loop.

$\rho_{\text{init}} \rightarrow \text{construct } V_{\text{eff}}[\rho] \rightarrow \text{solve KS eq.} \rightarrow \{\psi_k\} \rightarrow \rho_{\text{new}} = \sum_k |\psi_k|^2 \rightarrow \text{check convergence} \\ \rightarrow \rho_{\text{converged}}$

1.3 Molecular Dynamics (MD)

Starting point. Within the Born–Oppenheimer approximation the nuclei are treated as classical particles.

Equation: Newton’s equation of motion.

$$M_I \ddot{\mathbf{R}}_I = -\nabla_{\mathbf{R}_I} V(\{\mathbf{R}\}) \equiv \mathbf{F}_I. \quad (3)$$

The function $V(\{\mathbf{R}\})$ is called the *interatomic potential* or *potential energy surface (PES)*. In chemistry and the MD community it is commonly referred to as the *force field*, because its ultimate purpose is to compute forces $\mathbf{F}_I$.

Approximation. A classical force field $V(\{\mathbf{R}\})$ — such as the Lennard-Jones potential, Embedded Atom Method (EAM), CHARMM, or AMBER — is a *hand-crafted analytic function* with fixed functional form and parameters fitted from experiment or DFT data.

Time-integration loop (Verlet algorithm).

MD loop

Initial conditions: $\{\mathbf{R}_I(0), \mathbf{v}_I(0)\}$

↓

Compute forces: $\mathbf{F}_I = -\nabla V_{\text{FF}}(\{\mathbf{R}_I\})$ (analytic force field, very fast)

↓

Newton integration (Verlet):

$$\mathbf{R}_I(t+\Delta t) = 2\mathbf{R}_I(t) - \mathbf{R}_I(t-\Delta t) + \frac{\mathbf{F}_I}{M_I} \Delta t^2$$

↓

Update positions and velocities

↓

Return to “Compute forces”

The MD workflow is illustrated schematically in fig. 1. table 1 summarises the key differences between DFT and classical MD.

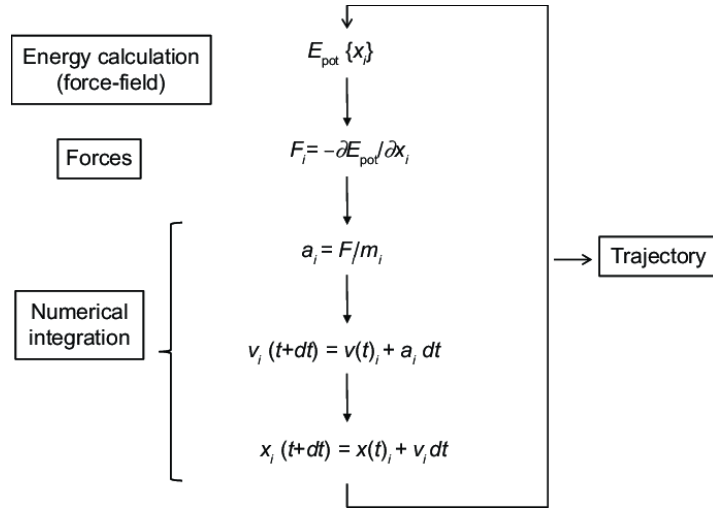


Figure 1: Schematic of the MD simulation workflow.

1.4 Ab Initio Molecular Dynamics (Ab Initio MD, AIMD)

In AIMD every MD time step requires a full DFT calculation.

Hellmann–Feynman theorem. The DFT total energy is a function of atomic positions, $E_{\text{DFT}} = E_{\text{DFT}}(\{\mathbf{R}_I\})$. The force on atom I follows from the converged electron density:

$$\mathbf{F}_I = -\frac{\partial E_{\text{DFT}}}{\partial \mathbf{R}_I}. \quad (4)$$

Table 1: Key differences between MD and DFT.

Feature	Molecular Dynamics (MD)	Density Functional Theory (DFT)
Physics basis	Classical mechanics (Newton’s laws)	Quantum mechanics (Schrödinger eq.)
Core variable	Positions and velocities of atoms	Electron density $\rho(\mathbf{r})$
Speed / Scale	ns– μ s, thousands of atoms	ps, tens to hundreds of atoms
Accuracy	Lower; requires predefined force fields	Higher; first-principles (ab initio)
Use cases	Long-timescale dynamics, diffusion	Band gaps, reactions, electronic structure

Workflow. Fix current atomic positions $\rightarrow$ run DFT to obtain electronic structure $\rightarrow$ compute atomic forces from the electronic structure $\rightarrow$ update atomic positions $\rightarrow$ repeat.

AIMD loop

Initial conditions: $\{\mathbf{R}_I(0), \mathbf{v}_I(0)\}$

$\downarrow$

With current $\{\mathbf{R}_I(t)\}$, run a full DFT SCF:

$\rho_{\text{init}} \rightarrow V_{\text{eff}}[\rho] \rightarrow \text{KS eq.} \rightarrow \{\psi_k\} \rightarrow \rho_{\text{new}} \rightarrow \dots \rightarrow \rho_{\text{conv}}$

Output: $E_{\text{DFT}}(\{\mathbf{R}_I\})$ and $\mathbf{F}_I = -\partial E_{\text{DFT}}/\partial \mathbf{R}_I$ (slow — full SCF each step)

$\downarrow$

Newton integration (identical to classical MD)

$\downarrow$

Update positions and velocities

$\downarrow$

Return to “run DFT”

1.5 Differences Between Methods

One of the largest differences between simulation methods in materials science is the trade-off between *computational cost* and *simulation scale*.

Density Functional Theory (DFT). Although DFT reduces the $3N$ -dimensional wavefunction $\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$ to a three-dimensional electron density $\rho(\mathbf{r})$, the Kohn–Sham formulation still requires solving a large set of single-particle equations $\hat{H}\psi_i = \varepsilon_i\psi_i$, i.e. diagonalising an $M \times M$ matrix where M is proportional to the number of atoms. The computational cost scales as $O(M^3)$. In practice, DFT can handle $\sim 10^2$ atoms, suitable for unit cells, surface slabs, or defect regions.

Molecular Dynamics (MD). Classical MD does not solve the electronic structure; instead it uses an empirical force field whose cost scales as $O(N)$ or $O(N \log N)$ (the latter when long-range Coulomb forces are treated with Ewald summation or SPME). This enables large length scales and long time scales, at the cost of reduced accuracy: force fields cannot reliably describe bond breaking, charge transfer, or other quantum effects.

Ab Initio Molecular Dynamics (AIMD). Because the atomic vibration time scale is ~ 1 fs, the MD time step must be $\Delta t \sim 1$ fs $= 10^{-15}$ s. A single DFT calculation can take minutes to hours, so AIMD is typically limited to $\sim$ ps ($\approx 10^3$ steps).

Total computational cost.

$$\text{Total cost} = \underbrace{\text{Cost per step}}_{\propto \text{system size}} \times \underbrace{\text{Number of steps}}_{\propto \text{simulated time}}. \quad (5)$$

This is why DFT/AIMD is regarded as “high-accuracy, small-scale, short-time” and classical MD as “low-accuracy, large-scale, long-time”.

Bridge: Machine-Learned Interatomic Potential (MLIP). To overcome the limitations of both approaches, Chapter 2 introduces MLIPs. Unlike classical analytic force fields, MLIPs use machine learning to encode DFT-quality potential energy surfaces: DFT training data is generated first, then an ML model is trained to approximate that surface. The result retains DFT accuracy while running at speeds close to classical force fields.

The scales accessible to each method are compared in fig. 2. table 2 gives a concise numerical overview.

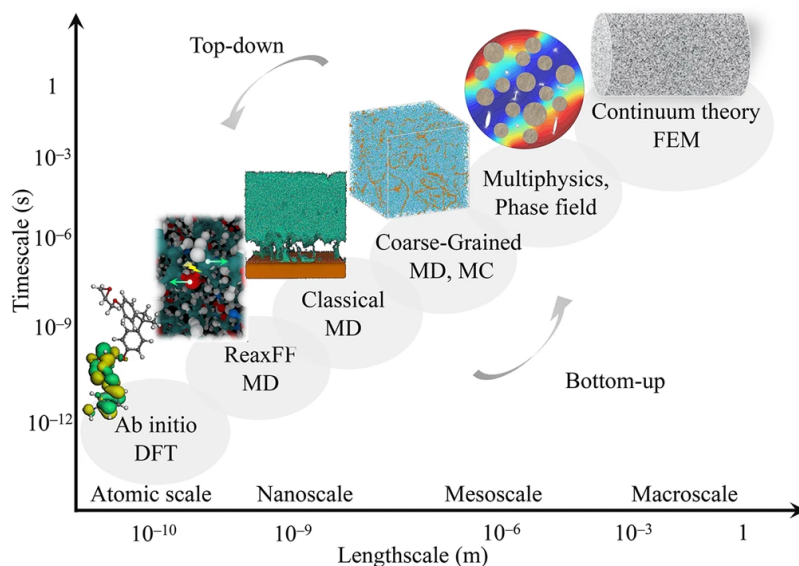


Figure 2: Simulation methods in the length-scale vs. time-scale plane. *Ab initio* DFT/AIMD occupies the bottom-left (high accuracy, small scale); classical MD occupies the top-right (fast, large scale). MLIPs aim to combine DFT accuracy with classical MD speed. Figure adapted from *Commun. Mater.* **4**, 66 (2023).

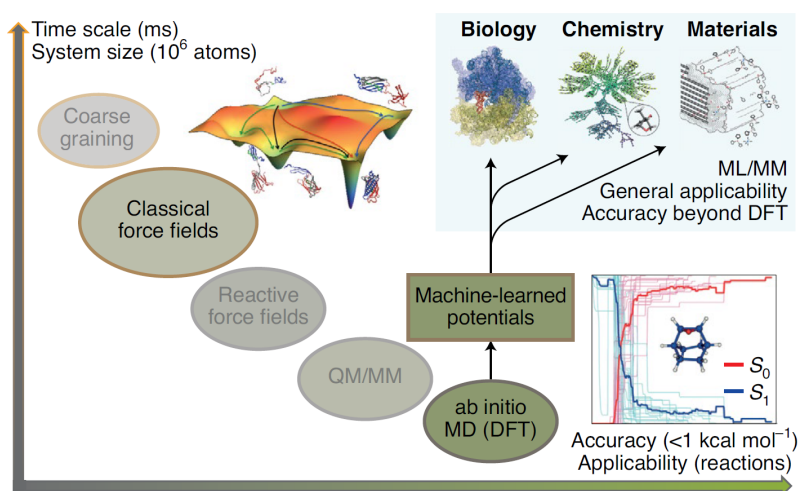


Figure 3: ML potentials as a potential solution to the trade-off between cost and accuracy of conventional atomistic simulations. Figure from *Nature Mater.* **20** (2021).

Table 2: Comparison of simulation methods.

Method	Time scale	Atom count	Accuracy
DFT (AIMD)	$\sim\text{ps}$	~ 100	High
Classical MD	$\text{ns}-\mu\text{s}$	$\sim 10^6$	Low-medium
MLIP-MD	$\text{ns}-\mu\text{s}$	10^4-10^6	Near DFT

1.6 Supplement: Comparison with DFT/MD/DMFT

1.6.1 DFT+DMFT and AIMD: differences

DFT+DMFT charge self-consistency loop. The two methods communicate through the *electron density* ρ , iterating in the space of electronic degrees of freedom:

$$\rho_{\text{init}} \rightarrow H_{\text{DFT}}^k \rightarrow \Sigma(i\omega_n) [\text{DMFT}] \rightarrow \rho_{\text{DMFT}} \rightarrow H_{\text{DFT}}^k [\text{update}] \rightarrow \dots \quad (6)$$

AIMD loop. The two methods communicate through *atomic positions* $\mathbf{R}_I$ and *forces* $\mathbf{F}_I$ — MD passes positions to DFT, DFT returns forces and energy to MD — iterating in the space of nuclear degrees of freedom:

$$\{\mathbf{R}_I(t), \dot{\mathbf{R}}_I(t)\} \rightarrow E_{\text{DFT}}(\{\mathbf{R}_I\}), \mathbf{F}_I [\text{DFT}] \rightarrow \mathbf{R}_I(t+\Delta t) [\text{Newton}] \rightarrow \dots \quad (7)$$

The analogy is summarised in table 3.

Table 3: DFT $\rightarrow$ DMFT vs. DFT $\rightarrow$ MD: structural analogy.

	DFT $\rightarrow$ DMFT	DFT $\rightarrow$ MD
Role of DFT	Band structure / Wannier Hamiltonian	PES / forces as reference
Reason DFT fails	Strong local correlations (Mott problem)	Too slow for large-scale dynamics
Core problem	How do electrons behave under correlations?	How do atoms move in time?
Equation	KS eq. $\rightarrow$ Dyson eq. + self-consistency	Newton: $M\ddot{\mathbf{R}} = -\nabla_R E$
Solved quantity	Green's function $G(i\omega_n)$	Atomic trajectories $\{\mathbf{R}_i(t)\}$
Connection	$\varepsilon_k \rightarrow \Delta(i\omega_n) \rightarrow \mathcal{G}^0$	Forces $\mathbf{F} = -\nabla E \rightarrow \mathbf{R}(t+\Delta t)$

1.6.2 Differences and limitations of DFT / MD / DMFT

A full comparison of all five methods is given in table 4.

Key message: DFT+DMFT extends DFT in the *energy* dimension (electronic correlations); MD extends DFT in the *time and length-scale* dimension. MLIP is the bridge that makes DFT-quality MD feasible.

1.6.3 Toy Model

table 5 draws the analogy between the Hubbard model used in the strongly-correlated context and the MD toy model used in the exercises (see section A).

1.6.4 Research story so far

Table 4: Full comparison of DFT, DFT+DMFT, classical MD, AIMD, and MLIP.

	DFT	DFT+DMFT	Classical MD	AIMD	MLIP
Object	Elec. gas	Correlated e^- dyn.	Nuclear motion	Nuclear motion	Nuclear motion
Core eq.	KS	Dyson+SC	Newton	Newton	Newton
Solved qty	ρ, ε_k	G, Σ	$\mathbf{R}_I, \mathbf{v}_I$	$\mathbf{R}_I, \mathbf{v}_I$	$\mathbf{R}_I, \mathbf{v}_I$
DFT input	—	ε_k	$F = -\nabla E$	$F = -\nabla E$	$F = -\nabla E$
Limitation	—	Ignores nonlocal corr.	Low accuracy	High CPU cost	Extrapolation
Solution	—	cDMFT, D Γ A	MLIP	MLIP	Active learning

Table 5: Analogy between the Hubbard model and the MD toy model.

	Hubbard model	MD + LJ/EMT
Equation	$H = -t \sum c^\dagger c + U \sum n_\uparrow n_\downarrow$	Newton: $F = ma$; LJ: $V(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$
Parameters	t (hopping), U (on-site repulsion)	ε (well depth), σ (atomic size)
Classic output	Mott transition (gap opens as U grows)	Energy conservation (NVE) + temperature (NVT)

Learning path

1. DFT introduced (done)
2. DFT success: band structure, metal/insulator prediction (done)
3. Two extensions of DFT:
 - Path A: strong correlations $\rightarrow$ DMFT (done)
 - Path B: large-scale dynamics $\rightarrow$ MD
4. MD equation of motion (Newton) — analogous to Dyson equation
5. Bottleneck of MD: how to compute $E(\{R\})$?
 - AIMD: accurate but slow/classical force field: fast but inaccurate
6. MLIP: trained on DFT data $\rightarrow$ best of both worlds

The two research directions from the author’s background are compared in table 6.

Table 6: Comparison of the two research directions.

	DFT+DMFT (previous work)	DFT $\rightarrow$ MLIP $\rightarrow$ MD (new direction)
Starting point	DFT	DFT
Problem solved	Electronic correlation (small scale)	Large-scale atomic dynamics
Role of ML	None	Fit PES, replace direct calculation
Computational bottleneck	QMC sampling	DFT training data generation

2 Machine-Learned Interatomic Potentials (MLIP)

2.1 Brief History of MLIP

In 2007, Behler and Parrinello proposed the following idea:

Use DFT to compute energies and forces for a large number of atomic configurations as training data, then train a machine-learning model to learn the mapping *atomic arrangement* $\rightarrow$ *energy/forces*.

The resulting model is called a **Machine-Learned Interatomic Potential (MLIP)**. It is a type of interatomic potential trained via ML; its accuracy is close to DFT and its computational speed is close to classical potentials, making it suitable for large-scale MD simulations with codes such as LAMMPS.

Review papers: Friederich *et al.*, *Nat. Mater.* **20**, 750 (2021); Behler, *J. Chem. Phys.* **145**, 170901 (2016).

The family tree of interatomic potentials is:

```
Interatomic Potentials (general)
|
|- Classical empirical: Lennard-Jones, EAM, ... (fast, less accurate)
|
'- MLIP
  |
  |- Behler-Parrinello Neural Network Potential (2007)
  |- Gaussian Approximation Potential (GAP)
  |- ACE: Atomic Cluster Expansion (2019)
  '- GRACE: ACE + Graph Neural Network (state of the art)
```

Atomic Cluster Expansion (ACE). ACE is a mathematical framework for describing the local atomic environment, proposed by Drautz in 2019. Its core idea is to expand the local chemical environment of each atom in a complete set of cluster basis functions — analogous to a Fourier expansion for periodic functions. This makes ACE (i) mathematically complete (no in-principle accuracy ceiling), (ii) systematically improvable by retaining higher-order terms, and (iii) more physically interpretable than Behler-type neural network potentials.

Graph Atomic Cluster Expansion (GRACE). GRACE extends ACE with a graph neural network architecture, enabling more efficient capture of long-range interactions and complex many-body effects. It is among the most advanced MLIP methods currently available.

2.2 Workflow

MLIP-MD workflow

— Training phase (offline) —

Use DFT to compute $\{E_{\text{DFT}}(\{\mathbf{R}_I\}), \mathbf{F}_I\}$ for a large and diverse set of configurations.

Train ML model: $V_{\text{ML}}(\{\mathbf{R}\}) \approx E_{\text{DFT}}(\{\mathbf{R}\})$.

— Inference phase (MD simulation) —

Initial conditions: $\{\mathbf{R}_I(0), \mathbf{v}_I(0)\}$

↓

Compute forces: $\mathbf{F}_I = -\nabla V_{\text{ML}}(\{\mathbf{R}\})$ (ML evaluation, orders of magnitude faster than DFT)

↓

Newton integration (identical to classical MD)

↓

Repeat

2.3 Differences from Other Computational Methods

Two key properties distinguish MLIP-MD from the alternatives:

1. Its scale is not limited by DFT cost.
2. Within the configuration space covered by the training data, its accuracy is close to DFT. If the simulation enters configurations not seen during training (e.g. extreme temperatures or novel chemical environments), MLIP can become inaccurate. This is the principal current limitation of MLIPs and the motivation for active learning.

table 7 compares AIMD and MLIP-MD directly.

Table 7: AIMD vs. MLIP-MD.

	AIMD	MLIP-MD
Source of forces	DFT computed on-the-fly	ML model prediction
Role of DFT	Online (each step)	Offline (training data only)
DFT running during MD?	Yes	No
Scale limitation	Limited by DFT cost	Not limited

The MLIP logic: first use AIMD (or static DFT on many configurations) to generate training data, then encode this DFT “knowledge” into the ML model; thereafter, DFT is no longer called during simulation.

2.4 Active Learning for Force Fields

Principal limitation of MLIP: poor extrapolation. If during an MD run atoms enter a configuration not covered by the training set (e.g. very short interatomic distances at extreme temperatures), the MLIP produces unreliable forces that can cause the simulation to crash.

Solution: uncertainty estimation and active learning. Modern MLIP frameworks (MACE, GRACE, GAP) include **uncertainty estimation**. When the model detects high prediction uncertainty for the current structure, it triggers an *active learning* loop:

1. Flag the uncertain structure.
2. Compute its energy and forces with DFT.
3. Add the new data point to the training set.
4. Retrain or fine-tune the MLIP.

This is conceptually equivalent to the self-consistency loop in DFT+DMFT: a feedback mechanism that drives the model towards convergence.

Part II

Exercise

A Toy Model: MD with LJ/EMT

Purpose. The LJ potential describes the potential energy between pairs of atoms. An MD simulation *animates* this potential: at each time step the force $F = -dV/dr$ is computed from the potential, and Newton's second law $F = ma$ updates each atom's position and velocity.

Tool: ASE (Atomic Simulation Environment). ASE is a Python package for atomic-scale simulations; it is pure Python, easy to use, and well-suited to small exercises. Other MD codes include LAMMPS, GROMACS, etc. For the interatomic potential we use **Effective Medium Theory (EMT)**, a slightly more accurate empirical potential than LJ that is built into ASE and supports copper (Cu) and other metals without requiring any DFT.

Ensembles. NVE and NVT are abbreviations for thermodynamic ensembles:

- **NVE:** total energy = constant, temperature fluctuates freely. NVE is an isolated system with no external perturbation.
- **NVT** (canonical ensemble): temperature = constant, total energy fluctuates. NVT is realised via a **Langevin thermostat**: a friction force and a stochastic force are added to each atom's equation of motion, coupling the system to a heat bath. In essence, these terms satisfy the *fluctuation-dissipation theorem*, ensuring the phase-space distribution follows the Boltzmann distribution.

The ensemble definitions are summarised in tables 8 and 9. tables 10 and 11 compare the LJ and EMT potentials and draw the analogy with the Hubbard model.

A.1 Lennard-Jones Potential: the simplest interatomic model

In MD simulations the interaction between atoms comes from the *interatomic potential*. The simplest model is the Lennard-Jones (LJ) potential:

$$V(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]. \quad (8)$$

The physical meaning of the parameters is given in table 12, and the effect of varying ε and σ is shown in fig. 4.

Physical origin of the two terms.

- **Repulsive term** $(\sigma/r)^{12}$: Pauli exclusion principle — rapid repulsion when electron clouds overlap.
- **Attractive term** $(\sigma/r)^6$: van der Waals attraction (London dispersion forces).

A.2 Physical Observables

A.2.1 Energy and Temperature

The core output of MD is **atomic positions and velocities**. The physical observables we record are derived quantities:

Position $\rightarrow$ potential energy

Table 8: Ensemble comparison.

Ensemble	Fixed quantities	Use case
NVE	N, V, E_{tot}	Verify numerical correctness (energy conservation)
NVT	N, V, T	Simulate realistic constant-temperature conditions

Table 9: Meaning of ensemble letters.

Letter	Meaning	Fixed quantity
N	Number of particles	Atom count constant
V	Volume	Volume constant
E	Energy	Total energy constant (NVE)
T	Temperature	Temperature constant (NVT)

Table 10: LJ potential vs. EMT.

	LJ Potential	EMT
Purpose	Conceptual demonstration, simplest model	Empirical potential for real metal simulations
Formula	Analytic expression, can be plotted	No simple formula; numerical method
Applicability	Noble gases (Ar, Ne)	Metals (Cu, Au, Ni, ...)
Analogy	Like the Hubbard model	Like a more complete multi-orbital model

Table 11: Analogy between Hubbard model and MD + LJ/EMT.

	Hubbard model	MD + LJ/EMT
Equation	$H = -t \sum c^\dagger c + U \sum n_\uparrow n_\downarrow$	Newton: $F = ma$; LJ: $V(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$
Diagram	t (hopping) and U (on-site repulsion)	r_{min} , ε , repulsive and attractive terms
Classic output	Mott transition (gap opens as U grows)	Energy conservation (NVE)temperature stabilisation (NVT)

Table 12: Physical meaning of LJ parameters.

Para.	Physical meaning	Effect on curve
ϵ	Well depth; attraction strength	Larger $\epsilon \Rightarrow$ deeper well $\Rightarrow$ stronger binding
σ	Effective atomic diameter ($V = 0$ distance)	Larger $\sigma \Rightarrow$ curve shifts right $\Rightarrow$ larger equilibrium distance

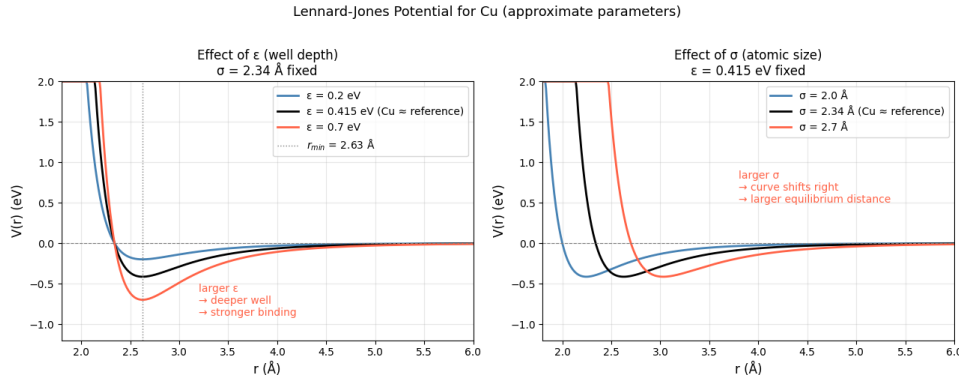


Figure 4: LJ potential curves as ϵ and σ are varied.

Velocity $\rightarrow$ kinetic energy
Velocity (statistical average) $\rightarrow$ temperature

Temperature is defined in statistical mechanics as the average kinetic energy of all atoms and is therefore an *output*, not an input. In ASE: `atoms_nve.get_temperature()`.

The simulation results for Cu using EMT are shown in fig. 5. The figure contains three panels, described in table 13.

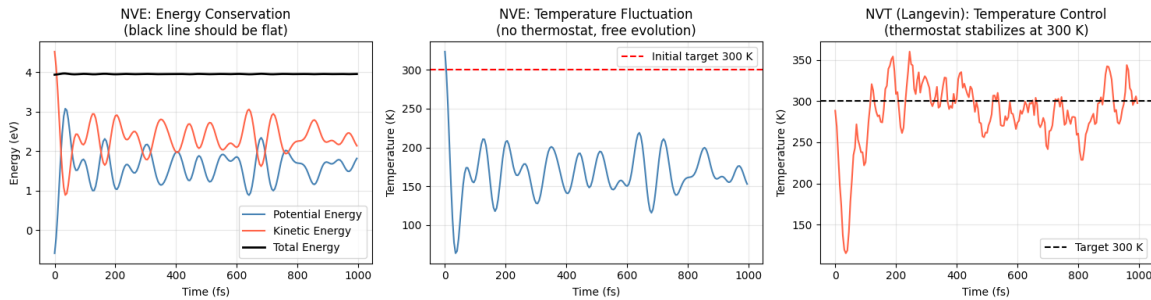


Figure 5: Cu (FCC) simulation using EMT as the interatomic potential. *Left*: NVE total energy. *Centre*: NVE temperature. *Right*: NVT temperature.

1. **NVE**: The system relaxes from a non-equilibrium initial state; energy exchanges between kinetic and potential forms, temperature fluctuates freely, and eventually stabilises (equipartition theorem).
2. **NVT (Langevin)**: A “virtual heat bath” controls the temperature. The Langevin thermostat adds (i) a friction force (slows over-fast atoms) and (ii) a stochastic force (mimics thermal fluctuations). Together they couple the system to a 300 K bath and force the temperature to converge to the target.

Analogy with DMFT: NVE is like running QMC without self-consistency; NVT is like the self-consistency loop that forces the system to converge to the target state.

Table 13: Description of the three panels in fig. 5.

Panel	Expected behaviour
Left: NVE energy	Total energy is nearly flat — proof of energy conservation
Centre: NVE temperature	Fluctuates around $\approx T_0/2$ without control
Right: NVT temperature	Gradually stabilises at $T_0 = 300$ K

A.2.2 Radial Distribution Function (RDF)

The RDF $g(r)$ is defined as the probability density of finding an atom at distance r from a reference atom, normalised to the uniform-density value:

$$g(r) = \frac{V}{N^2} \left\langle \sum_{i \neq j} \delta(r - r_{ij}) \right\rangle. \quad (9)$$

Its characteristic features are:

1. **Peak position:** determined by the neighbour distance.
2. **Peak width:** broadened by thermal fluctuations.
3. **Peak height:** proportional to the coordination number (number of atoms at that distance).

The RDF signatures for different states of matter are summarised in table 14. The FCC nearest-neighbour shells for Cu are listed in table 15.

Table 14: RDF signatures for different physical states.

State	RDF appearance
Perfect crystal (0 K)	Sharp δ -function peaks
Solid (finite T)	Broadened Gaussian peaks at neighbour distances
Liquid	Prominent first peak, rapid decay to $g(r) \rightarrow 1$
Gas	Nearly featureless, $g(r) \approx 1$

Table 15: FCC nearest-neighbour shells for Cu ($a \approx 3.60$ Å).

Shell	Distance	Coordination number	RDF peak height
1st	$a/\sqrt{2} \approx 2.55$ Å	12	Highest
2nd	$a \approx 3.60$ Å	6	Lower
3rd	$a\sqrt{3/2} \approx 4.41$ Å	24	Higher than 2nd
4th	$a\sqrt{2} \approx 5.09$ Å	12	—

The RDF of Cu FCC at 300 K is shown in fig. 6. At finite temperature, the perfect-crystal δ -function peaks broaden into Gaussian peaks due to thermal vibrations.

For the Ar phase transition (LJ potential, varying temperature), the RDF in fig. 7 shows the progression from solid to liquid. EMT is specific to metals and cannot describe phase transitions, so LJ is used here instead.

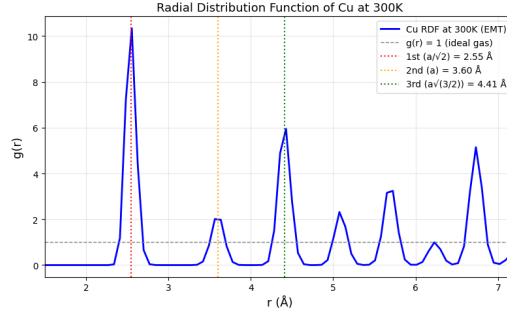


Figure 6: RDF of Cu FCC at 300 K (EMT). The first peak at $\approx 2.55 \text{ \AA}$ corresponds to 12 nearest neighbours; thermal broadening produces Gaussian peaks.

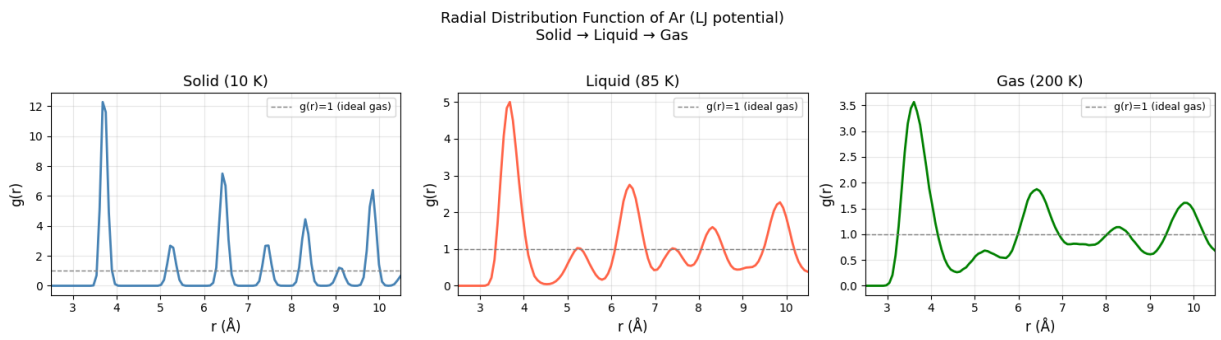


Figure 7: RDF of Ar at different temperatures (LJ potential), illustrating the solid-to-liquid phase transition.

B Toy Model: MD with MACE

Instead of EMT, one can use **MACE**, a pre-trained MLIP, by installing **mace-torch**:

```
pip install mace-torch
```

It is important to note that MACE is not a numerical simulation result but a **neural network fitted to a DFT potential energy surface**: DFT was run on millions of structures to obtain energies and forces, and the network learned the mapping *structure* $\rightarrow$ *energy/force*. Once trained, DFT is no longer needed.

table 16 compares the three interatomic potentials.

Table 16: Comparison of LJ, EMT, and MACE as interatomic potentials.

	LJ	EMT	MACE
Form	Analytic formula	Semi-empirical numerical	Neural network (numerical)
Parameter source	Manual tuning	Fitted to experimental data	Fitted to DFT results
Accuracy	Low (noble gases)	Medium (metals, qualitative)	High (near DFT)
Speed	Very fast	Fast	Slow (but far faster than DFT)

The MACE results for Cu are shown in figs. 8 and 9.

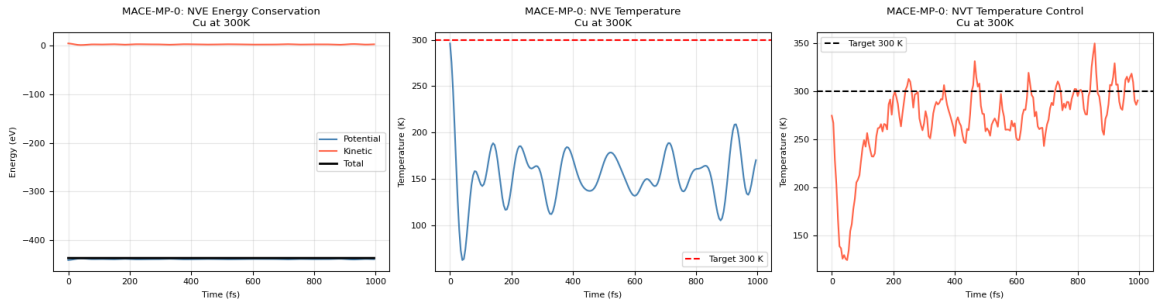


Figure 8: Cu simulation using MACE. *Left*: NVE energy. *Centre*: NVE temperature. *Right*: NVT temperature.

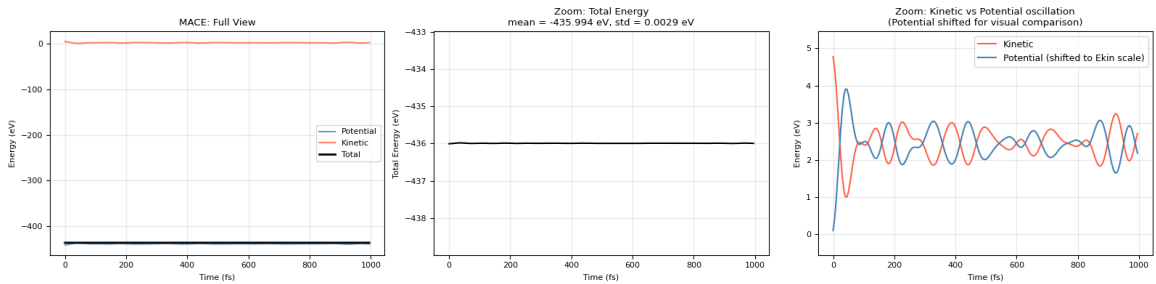


Figure 9: Zoom-in of the MACE Cu results.

Q1: Why is the energy so different from EMT? (+3 eV vs. -430 eV) This is the most fundamental question, and the answer is: **absolute energy values have no physical meaning; only energy differences do**. The EMT energy reference is arbitrary (e.g. isolated atom = 0). MACE is trained on DFT data with a physically meaningful reference (true atomic energies), so the total energy is large and negative because atoms release energy when they bind together.

Analogy: just as in DMFT the absolute value of the self-energy depends on the double-counting correction, the physical content lies in the frequency-dependent structure, not the absolute value.

Table 17: Energy reference comparison between EMT and MACE-MP-0 for 108 Cu atoms.

	EMT	MACE-MP-0
Energy reference	Arbitrary (isolated atom = 0)	DFT absolute energy
Total energy (108 Cu)	$\approx +3\text{ eV}$	$\approx -430\text{ eV}$
Cohesive energy per atom	Inaccurate	$\approx -4\text{ eV/atom}$ ($\approx \text{exp. } -3.5\text{ eV}$)

Q2: Why do the temperature plots have similar shapes but different values? Both plots show temperature dropping from near 300 K as the system relaxes from a non-equilibrium initial state. The difference is that MACE describes a stiffer Cu (larger force constants) than EMT, leading to slightly different oscillation frequencies and equilibrium temperatures.

C Next Steps in Research

The appropriate next step depends on the research objective:

1. **Use a pre-trained model (as in the exercise above).** Suitable for general materials, rapid validation, and industrial applications. MACE-MP-0 provides reasonable accuracy for most materials.
2. **Fine-tune or train from scratch.** When precise properties of a specific material are needed (e.g. a new oxide, phase transitions, defect behaviour), the pre-trained model may not be accurate enough. In that case one generates DFT training data for the target material and trains a dedicated MLIP.

Practical example. The workflow in Behler’s group is an instance of the second approach: they develop the methodology for Neural Network Potentials (NNPs), studying how to generate training data more efficiently, design better descriptors, and improve generalisation. A typical project involves:

1. Run DFT (VASP or Quantum ESPRESSO) on a batch of structures.
2. Train an NNP on these data (using their `n2p2` tool).
3. Run large-scale MD with the trained NNP.
4. Analyse MD output (RDF, diffusion coefficient, phase transitions, etc.).

The toy-model exercise in sections A and B already covers steps 3 and 4 conceptually.

References

- [1] P. Hohenberg and W. Kohn, *Inhomogeneous electron gas*, Phys. Rev. **136**, B864 (1964).
- [2] W. Kohn and L. J. Sham, *Self-consistent equations including exchange and correlation effects*, Phys. Rev. **140**, A1133 (1965).
- [3] B. J. Alder and T. E. Wainwright, *Phase transition for a hard sphere system*, J. Chem. Phys. **27**, 1208 (1957).
- [4] A. Rahman, *Correlations in the motion of atoms in liquid argon*, Phys. Rev. **136**, A405 (1964).
- [5] J. Behler and M. Parrinello, *Generalized neural-network representation of high-dimensional potential-energy surfaces*, Phys. Rev. Lett. **98**, 146401 (2007).
- [6] R. Drautz, *Atomic cluster expansion for accurate and transferable interatomic potentials*, Phys. Rev. B **99**, 014104 (2019).
- [7] P. Friederich *et al.*, *Machine-learned potentials for next-generation matter simulations*, Nat. Mater. **20**, 750 (2021).
- [8] J. Behler, *Perspective: Machine learning potentials for atomistic simulations*, J. Chem. Phys. **145**, 170901 (2016).